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Topics in Medicine and Surgery: Review

How to Assess Stress in Reptiles

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Abstract

Stress, as an adaptive response of any animal to a stimulus which presents a threat to homeostasis, can occur in reptiles. Yet many veterinarians fail to recognize signs of stress in reptiles. In this article, evaluation of reptile stress is discussed with a focus on the analysis of behavioral changes, neuro-endocrine correlation, and biochemical and physiological effects in reptiles. If stress can be assessed and treated properly in captive reptiles, their health and well-being can be optimized.

***Key words:* animal welfare; captivity; neuro-endocrine; reptiles; stress**

Stress can be defined as a normal biologically adaptive response of an individual to internal or external stimuli representing a threat to homeostasis.¹ Four main factors are found in the response to stress: 1) behavior, 2) alterations in the functioning of the autonomic nervous system, 3) neuro-endocrine responses, and 4) immune responses. These four elements are correlated and coordinated by the hypothalamus and corticotrophin-releasing hormone (CRH). Knowledge of all of the elements of stress delineated above can be useful in efforts to measure this disease condition in reptiles. Stress does not only induce activation of the sympathetic nervous system but also directs a neuroendocrine response, initially slower but with potentially longer lasting effects derived from activation of the hypothalamus/pituitary/adrenal (HPA) axis. This neuro-endocrine axis is essential in the regulation of physiological functions (e.g., immune competence, reproduction, metabolism, behavior). Activation of the HPA axis is initiated when the CRH produced by the hypothalamus is released following a threat, stimulating the pituitary to secrete adrenocorticotrophic hormone (ACTH). The ACTH acts on the adrenal cortex to stimulate the production of glucocorticoids. The name suprarenal is not used in herpetology since this gland is not present dorsal to the kidneys in these more primitive vertebrates. The adrenal gland does not undergo embryonic migration and remains very close to the renal vein and the gonad. The adrenal gland consists of two tissue types: tissue of mesodermal origin (steroidogenic inter-renal cells) and those compounded with cells rich in fat. Functionally similar to cells of the mammals' adrenal cortex, secretion from the reptile adrenal gland consists mainly of corticosteroids. Chromatin tissue derived from the neural crest is colored intensely by histologic dyes because the cells have a high secretory component of both

adrenaline and noradrenaline. This is the evolutionary equivalent to the adrenal medulla of mammals. Unlike mammals, these cell types are mixed in a common matrix and do not form distinct regions. Although baseline cortisol levels are identifiable in blood, the main glucocorticoid in reptiles is corticosterone. Capture and events surrounding this process are known to be one of the main stressors in reptile species.² This fact should not be overlooked, but many other conditions affect corticosteroid release. Release of corticosteroids affects different reptile species over different time periods; therefore, all undesirable effects of these internally produced compounds must be considered in captive reptile species (Table 1).

Stress Behavioral Measurement

Behavioral adaption is considered the optimal response by the animal to an external stressor. Avoiding a threat (i.e., derived from natural behavior of evading predators) is best and biologically requires the least effort to survive or avoid an unfavorable situation. When the individual is able to safely increase the distance from the stressor, the stress response does not usually affect the animal's long term welfare. However, if the reptile is not able to respond in an appropriate manner, the stressor may cause negative effects on the animal's welfare, defined as *suffering*. Escape, however, is not always possible, particularly for captive animals living in a confined area. The behavior shown by animals during the response to stress may provide a measurement of their inner or subjective state and may be useful as a measurement of the stress. If the stressor persists, the individual may show abnormal nonfunctional behaviors. These "displaced" behaviors may help the animal face the psychological aspect of the stressful event and also reduce the full effect of its physiologic response. Specific behavior parameters that are considered useful

indicators of chronic stress include: aggression, anorexia, redirected activities, stereotypies, and displacement behaviors.³ Variations in the deposition of carotenes have been reported in stressed lizards, with a corresponding change to their normal skin colors.⁴

Autonomous Nervous Measurement of Stress

Once an animal perceives a stimulus as a threat, its brain releases a neuro-endocrine response that consists of acute and chronic phases. The acute phase is regulated by the sympathetic nervous system, which is immediately activated, short-acting, and has a direct impact on most body tissues. Catecholamines are released from the adrenal medulla seconds after the perception of a triggering stimulus and their catabolism is rapid. The magnitude of this response can be determined by measuring serum catecholamine levels, glucose, heart rate, and blood pressure. However, determining these measurements may contribute to a stress response. Adrenocorticotrophic hormone, glucocorticoids, and glucose are physiologic parameters that can be measured to determine the level of stress in an individual animal. Hyperglycemia, increased hepatic glycogen, protein catabolism, and increased gluconeogenesis enzyme activity have also been identified in animals undergoing stressful events. However, certain studies show differing results as to the relationship of the physiologic parameters delineated above and stress. Moreover, there is no reasonable explanation to describe the significant effects of stress on blood glucose or hepatic glycogen levels. In part, these inconsistent results may represent the important correlation of low temperatures and seasonal variation on the reptiles' physiologic response. These variations are an excellent means to

assess the magnitude of an acute response to stress, but are not reliable when trying to determine a long-term response. This fact, together with the difficulties in measuring parameters such as heart rate under field conditions, make the response of the autonomous nervous system less valuable in the assessment of stress in reptiles that live in a captive setting.⁵ Reptiles which are progressively and chronically stressed may be predisposed to obesity and hepatic lipidosis. Conversely, acutely or intensely stressed reptile species are predisposed to emaciation, immune depression, and reproductive difficulty.

Neuro-Endocrine Measurement of Stress

Due to the activation of the HPA axis, glucocorticoids are released into the blood stream. Glucocorticoids enhance their effects through the alteration of circulating levels of sex steroids and other hormones. The objective of a stress response is to maximize energy availability to body systems in order to face the physiological challenge(s). Therefore, the stress response is a powerful inhibitor of non-essential systems that, although critical for long-term survival, are not essential for the day to day survival of the animal, (e.g., growth, reproduction).⁶ Corticosterone inhibits reproductive behaviors that require significant amounts of energy, such as aggression or territorial defense, although the occurrence of intercourse does not appear to be affected because it requires relatively low energy consumption. However, without support behaviors associated with reproduction among males, the chance of reproductive success is probably low.⁷ Marine turtles are able to reabsorb their eggs and nullify the reproductive drive when they suddenly encounter changes to their egg laying environment. The short-term catabolic effects of high

glucocorticoids levels may be beneficial (particularly when combined with behaviors that tend to enhance escape from a stressful situation), but if they remain high and it becomes chronic it could lead to muscle degeneration and growth suppression.⁶ Corticosterone levels in reptiles vary between males and females. Adult male alligators (*Alligator mississippiensis*) have significantly higher corticosterone levels than adult females in both captive and wild populations.⁷ The same has been reported in tortoises.⁷ These differences, however, were not found when comparing juvenile male and female loggerhead sea turtles (*Caretta caretta*) in Florida.⁷

Plasma corticosterone levels can also vary according to the degree of individual dominance. In subordinate animals, plasma corticosterone levels are significantly higher than in their dominant cohorts. Additionally, the presence of daily cycles and seasonality of plasma corticosterone levels have been reported in reptiles, with the peak recorded when the animals are most active during a 24 hour cycle. However, the daily cycle of the plasma corticosterone levels has been reported to be sensitive to housing conditions. According to one investigation, corticosterone levels in shed skin of the African house snake (*Lamprophis fuliginosus*) are reported to be associated with increased hormone levels in feces during the shedding process.⁷

Capture and restraint of reptiles has been found to increase stress-related corticosterone levels in most of the animals tested,² although this does not occur consistently in gopher tortoises (*Gopherus polyphemus*).⁸ Problems associated with stress related increases in reptile corticosterone levels can be attenuated by measuring the glucocorticoid levels in the animal's feces. Fecal sample collection does not require capture or containment of the animal; therefore, variations in secretion patterns are minimized. While corticosterone levels are usually determined

through blood sample measurements, fecal and skin samples may also be used to measure stress.^{9,10}

Hematological Measurement of Stress

The effect of the stress response on the immune system is complex; however, the response of the immune system during stress has typically been associated with the HPA axis response since increased glucocorticoid levels in blood typically causes immune suppression and reduction of antibody production. These changes minimize the use of energy and favor immediate activity that may be used for survival. However, immune system suppression also increases disease occurrence in affected reptiles. Increased glucocorticoids levels are associated with heterophilia and lymphopenia, and in some species can affect degranulation of the granulocytes (e.g., heterophils and eosinophils).^{11,12} The stress leukogram demonstrates the effect that stress induced increases in creatine phosphokinase have on diagnostic test results. Additionally, changes in cholesterol and triglycerides vary depending on the stressor, although generally these tend to decrease. Hyperglycemia has been reported in stressed crocodilians and ophidians,⁷ and increased hematocrit, recent thymic emigrants, sodium, chloride, and potassium levels have been reported in terrapins subjected to hyperosmotic stress.⁷ Acute-phase proteins, which are produced in response to inflammation and infection,¹³ have also been associated with the regulatory activity of the HPA axis during the neuro-endocrine response to stress.^{13,14} Acute-phase proteins are considered non-specific components of the innate immune system, being involved in restoration of homeostasis. Unfortunately,

this type of determination has not yet been standardized in reptiles; consequently, it is a new research field to develop.

Conclusions

Reptiles are subject to many of the same stressors encountered in higher vertebrates, and while it is common to think that reptiles are less likely to be negatively impacted by these stressors, it is untrue. To minimize the likelihood of stress causing negative physiologic effects on captive reptiles, it is important to understand the different clinical signs associated with stress and methods for measuring stress in these animals. It is only through raising our awareness of these potential issues that we can minimize the impacts of stress on captive reptiles.

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Table 1. Time to response (corticosterone release) after stressor.

<u>Stressor</u>	<u>Time</u>
Capture	3 minutes to 6 hours
Exposure to cold and heat	15 to 25 min
Hyperosmotic saline injection	30 minutes
Intermittent change of environment	30 minutes
Repeated blood sampling	1 to 4 hours
Severe restraint	4 to 8 hours
Hypotonic fluid injection	8 days
Exposure to salt water	1 to 4 weeks
Hierarchy with dominating males	10 to 30 days
Overcrowding	10 to 14 days
Low relative humidity	3 weeks

(*) *The time to response is deemed the time to stimulation of corticosteroids release*